

CY 001: Preparatory Chemistry - 1

Course Code	CY 001
Course Name	Preparatory Chemistry - 1
L-T-P-C	3-1-0-4
Intended for	Preparatory Students
Distribution	Core
Approval	50th BoA (Jul 2023)

Course Contents

Physical Chemistry

- **Kinetic Theory:** Idea of distribution function, properties of gamma functions; transformation properties for Cartesian to polar coordinates. Maxwell's speed and energy Distributions (derivations for 1, 2 and 3 dimensions); distribution curves; different types of speeds and their significance, frequency of collisions against a surface; frequency of collisions against a surface; frequency of binary collisions; mean free path
- **Thermodynamics:** System and Surroundings, walls; reversible and irreversible processes; isothermal, adiabatic and other processes; work, partial and total derivatives; exact differentials and state functions, definitions of thermodynamic functions: Zeroth Law (T), First law (U) and second law (S); other functions like H, A, and G. Carnot's cycle and theorems; changes of thermodynamic functions in irreversibility and entropy, importance of H in thermo-chemistry, Maxwell's relations.
- **Chemical Kinetics:** Order and molecularity of reactions, first and second order reactions, average life period, concept of Arrhenius activation energy

Inorganic Chemistry

- **Inorganic Chemistry:** Periodicity, general trends, blocks of periodic table, s-block, p-block, and introduction to f-block, VSEPR, valence band theory, electron deficient bonding, thermodynamics of reduction processes.

Organic Chemistry

- **Organic Chemistry:** Classification and nomenclature of organic compounds, hybridization, dipole moment and bond energy, factors influencing electron availability: inductive effect, electromeric effect, resonance, mesomeric effect or conjugative effect, hyperconjugative effect, steric effect, H-bonding force etc, concept of organic acid and base, substitution and elimination reactions.

Text Books

1. J. D. Lee, **Concise Inorganic Chemistry**, 5th Edition, ELBS, 1996.
2. R. T. Morrison and R.N. Boyd, **Organic Chemistry**, 5th Edition, Printice Hall of India Pvt. Ltd. 1990.
3. G. Solomons and C. Fryhle, **Organic Chemistry**, John Wiley and Sons (Asia) Pte Ltd.
4. D. A. McQuarrie, and J. D. Simons, **Physical Chemistry**, Viva Books, 1998.
5. Irving M. Klotz and Robert M. Rosenberg, **Chemical Thermodynamics: Basic Concepts and Methods**, Wiley, 2008.